D5 - Automated Analysis Platform (SPBM): User and Audit Guide

Stockpile reconstruction and block model platform, version 1.0.0

> **Reproducibility standard.** A mine-planning engineer who did not develop the project should be able to install the software, reproduce every number in the report from the raw files, and add new data, using only this guide.

1. What the platform does

`spbm` is a Python package with a command-line interface. One command imports FMS dump records and survey change solids, checks them, reconstructs material movement (CPST algorithm, D3), builds the block model, quantifies uncertainty, validates the result, and writes an auditable record of the run.

Task 8 requirement	Where it happens	Output
1. Import new FMS records	`io.scan_sources`, `io.read_fms`	`audit/data_registry.csv`
2. Import new survey surfaces	`io.read_dxf_triangles`, `dataset.load_dataset`	cached grids in `data/cache/`
3. Data-quality checks	`qaqc.run_qaqc`	`qaqc/*.csv`, `qaqc/fig_*.png`
4. New placement and modelling period	`io.survey_register`, `growth.temporal_alignment`, period assignment in `movement.prepare_loads`	`audit/survey_register.csv`, `growth/temporal_alignment.csv`
5. Material-movement algorithm	`movement.reconstruct`	`movement/load_destinations.csv`, `transport_plans.npz`
6. Update block model	`blockmodel.*`	`blockmodel/stockpile_block_model.csv`, `.vtk`
7. Tonnage and metal inventories	`blockmodel.summary_tables`, `validation.class_inventory`	`blockmodel/summary_*.csv`, `validation/inventory_by_class.csv`
8. Uncertainty	`uncertainty.monte_carlo`, `sensitivity_oat`	`uncertainty/*.csv`, block P10/P50/P90/CV
9. QA/QC and reconciliation outputs	`validation.*`	`validation/*.csv`, `growth/period_reconciliation.csv`
10. Visualisations and exports	`bmviz`, `growth`, `qaqc`	`*/fig_*.png`, `.vtk`, `.asc`, `report_tables.md`
11. Audit records	`audit.*`	`audit/run_manifest.json`, `run_log.csv`, `history/<run_id>/`, `change_report.md`

2. Installation

Requirements: Python ≥ 3.10 on Windows, macOS or Linux. No commercial software is needed. Results open in Excel (CSV), ParaView (VTK), any GIS (ESRI ASCII grids), and Vulcan/Datamine/Leapfrog (CSV block model with centroids and sizes).

```
git clone <repository>            # or unzip the submission
cd mine413
python -m venv .venv && source .venv/bin/activate     # Windows: .venv\Scripts\activate
pip install -r requirements.txt
python -m pytest -q               # 8 tests, about 10 s; must all pass
```

Dependencies (all open source): numpy, scipy, pandas, matplotlib, ezdxf, pyyaml, openpyxl, pytest. Versions used are recorded in every `run_manifest.json`.

3. Input formats

Input	Format	Requirements
FMS	CSV with columns `Cycle ID, Truck PayLoad, Au, Ag, Cu, Dens, X (Dump Event), Y (Dump Event), Z (Dump Event), Dump_tstamp` (timestamp `M/D/YYYY H:MM`)	Any number of files. Records are de-duplicated by Cycle ID (first occurrence kept, duplicates flagged)
Surveys	DXF with 3DFACE entities forming a closed change solid (Vulcan "STOCK_HGHS" export)	File name must contain the date as `YYMMDD` (e.g. `Topo_260402.dxf`). Wrong dates are corrected in `config/survey_overrides.csv`, not by renaming
Configuration	`config/project.yaml`	All parameters, each tagged with its assumption ID
Manual interventions	`config/survey_overrides.csv` (`file, assigned_date, survey_time, reason, approved_by`)	Every override needs a reason and an approver. Overrides are copied into each run manifest

Sources are listed in `inputs.sources` (default: the original `Data_Topo 2026.zip`, read in place, and the folder `data/inbox/`).

4. Operating instructions

4.1 Reproduce the submitted analysis

```
python -m spbm run                # full analysis, about 35 min (60 Monte Carlo realisations)
python -m spbm run --fast         # about 30 s: base reconstruction, block model, QA/QC, balances only
python -m spbm run --mc 200       # more realisations
```

Results are written to `outputs/`. The run is deterministic: the base case has no randomness, and the Monte Carlo uses the fixed seed in the configuration. Re-running reproduces identical numbers, which you can check by comparing `outputs/history/*/run_manifest.json`.

4.2 Routine update when new data arrive (Task 8)

1. Copy the new FMS export(s) and the new survey solid(s) into `data/inbox/`.
2. If a survey file name is wrong, add a row to `config/survey_overrides.csv` with the reason.
3. Run

   ```
   python -m spbm update
   ```

   The platform lists the files it has not seen before (by SHA-256), re-runs the complete chain, and writes `outputs/audit/change_report.md` comparing tonnes, metal and classes with the previous model version.
4. Review, in this order: `qaqc/fms_flag_summary.csv`, `qaqc/survey_qaqc.csv`, `growth/period_reconciliation.csv` (large unattributed or retained volumes mean the FMS and the survey disagree), `validation/geometry_check.csv`, then the block model.
5. Approve the model version by recording the `run_id` in the decision log. Loads after the latest survey are flagged *Provisional*. They will be reconciled automatically once the next survey is imported.

Why a full re-run instead of an incremental patch? A full run takes minutes, gives identical results for identical inputs, and lets late corrections (e.g. a re-dated survey or a corrected FMS record) propagate correctly through all later periods. Incremental state files would be faster but harder to audit.

4.3 Example execution: demonstrating an update

The repository includes a demonstration of the update workflow. A model is built with data up to 22 March, then updated with the remaining data (`examples/demo_update.sh`):

```
python -m spbm run --fast --until "2026-03-22 23:59" --out outputs_demo
python -m spbm run --fast --out outputs_demo
cat outputs_demo/audit/change_report.md
```

4.4 Other commands

```
python -m spbm qaqc      # data checks only, prints the flag summary
python -m spbm status    # registered input files and run log
python tools/make_diagrams.py   # redraw workflow/algorithm diagrams
python tools/build_docs.py      # rebuild DOCX/PDF documents and the presentation
```

5. Output reference

File	Content
`blockmodel/stockpile_block_model.csv`	One row per block containing material. See D3 Section 5 for fields
`blockmodel/stockpile_block_model.vtk`	Same, as a VTK structured grid (ParaView: *Threshold* on `tonnes > 0`)
`blockmodel/block_model_definition.json`	Origin, block size, units, CRS note
`blockmodel/summary_*.csv`	Inventory by information class, confidence, 4 m bench, material source; grade-tonnage
`movement/load_destinations.csv`	For each load: dump X/Y, volume placed, retained, destination centroid, mean push distance, spread
`movement/transport_plans.npz`	Transport matrix of every period (full traceability of the allocation)
`movement/*.asc`	Floor plane, pre-record surface, final model surface, latest surveyed surface (ESRI ASCII grids)
`qaqc/`	Records with flags, flag summary, daily coverage, grade signatures, survey QA, figures
`growth/`	Period reconciliation, temporal alignment, growth figures
`validation/`	Mass/metal balance, inventory by class, volume reconciliation, geometry, forward-model temporal validation, twin experiment, robustness
`uncertainty/`	Monte Carlo realisations and global summary, tornado table and figures
`report_tables.md`, `summary.json`	Key numbers for the report
`audit/`	Data registry, survey register, run manifest, configuration used, run log, change report
`history/<run_id>/`	Archived manifest and key tables for every model version

6. Audit and reproducibility controls

- **Input identity**: every input file is hashed (SHA-256) and registered once with the date it was first seen. The manifest lists the hashes used by the run.
- **Configuration identity**: the configuration is hashed and copied (`config_used.json`). Every parameter is linked to an assumption ID.
- **Code identity**: the Git commit and all library versions are recorded.
- **Manual interventions**: only through `survey_overrides.csv`, with a reason and an approver, copied into the manifest. Raw files are never edited.
- **No silent deletion**: FMS records are flagged, never removed. Exclusions (duplicates, missing fields, bad timestamps) are listed in the flag file.
- **Model versions**: `history/` keeps each version's manifest and key tables; `change_report.md` documents differences.
- **Verification**: `python -m pytest` checks rasterisation, the transport solver, column operations, the avalanche model, survey re-dating and end-to-end mass/metal conservation.

7. Usability for different levels of expertise

- **Planners / managers**: read `report_tables.md` and the PNG figures. Use the block model CSV in Excel or the mine-planning package. Filter by `confidence` and `info_class`.
- **Engineers**: change parameters in `config/project.yaml` (comments explain each one and link to D2) and re-run.
- **Developers**: modules are small and documented; the tests define the expected behaviour.

8. Known limitations of the platform

- Survey inputs must be closed change solids (the format supplied). Open surfaces are detected (QA reports `open_cells`) but are not yet converted automatically. Recommended extension: accept full-pile surfaces and compute the change solids internally.
- Single-stockpile configuration. Several stockpiles would need one configuration each.
- There is no graphical user interface (not required). Outputs are files and figures.